

NGO Volunteer App - Architecture Specification

1. Executive Summary

The Volunteer App has been overhauled from a standalone prototype into a professional coordination node. The system now follows a strict 'Frozen Schema' data contract, ensuring that every submission is immediately actionable by the NGO Dashboard's AI pipeline.

2. Database Architecture (Shared Collections)

Collection: volunteers

Operational data synced with NGO Dashboard. Fields: name, skill_type, availability_status, assigned_zone, current_latitude/longitude, utilization_status.

Collection: assignments

The bridge for mission allocation. Fields: assignment_status, assigned_volunteer_id, priority_rank, location_name, zone_label, accepted_at.

Collection: reports

The unified intake pipeline. Fields: source_type (volunteer), submitted_by_id, category, severity_level, description, zone_label.

3. Core Logic Patterns

- Transactional Claim System: Uses db.runTransaction to prevent double-booking.

- Zon

Generated by Antigravity AI Engineering Suite.